5 number summary and box plot

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Primary Concepts

i) Minimum

ii) First Quartile (25 percentile) Q_1 (Q_1)

iii) Median (Q_2) 50%

iv) Third Quartile (Q_3)

v) Maximum

Removing the outliers

$< -4.5 \quad \boxed{1\,2\,3\,5,\,6} \quad 15.5 >$

$x = \{1, 2, 2, 2, 3, 3, 4, 5, 5, 5, 6, 6, 6, 7, 8, 8, 9, 29\}$ → outliers

Low Fence <—> Higher Fence

Lower Fence $= Q_1 - 1.5(IQR)$ → Interquartile Range

Higher Fence $= Q_3 + 1.5(IQR)$

$Q_1 = 25 \text{ percentile} = \frac{25}{100}(19+1)$

$= \cancel{\frac{25}{100}} \times \cancel{20}$

$= 5^{th}$ Value

$\boxed{Q_1 = 3}$

$Q_3 = 75 \text{ percent} = \cancel{\frac{75}{100}}\ \cancel{(20)}$

$= 15^{th}$ Value

$\boxed{Q_3 = 8}$

$\boxed{IQR}$

$Q_3 - Q_1 =$

$IQR = Q_3 - Q_1$

$= 8 - 3$

$IQR = 5$

Lower Fence $= Q_1 - 1.5(IQR)$

$= 3 - 1.5(5)$

$= 3 - 7.5$

Lower Fence $= -4.5$

Higher Fence $= Q_3 + 1.5(IQR)$

$= 8 + 1.5(5)$

$= 8 + 7.5$

H.F $= 15.5$

[- 4.5, 15.5]

5 Number

min = 1

$Q_1 = 3$

Q_2 (Median) = 5

$Q_3 = 8$

Max = 9